

CARBON AND ITS COMPOUNDS

Student Learning Pack

1. Metadata

Topic	Carbon and its Compounds
Level	Class X
Lesson Length	60 minutes

Scope Note

Focus on carbon's tetravalency, catenation, saturated vs unsaturated compounds, hydrocarbon classification, basic properties and historical context.

2. Learning Objectives

- LO1: Explain how carbon's tetravalency and catenation enable it to form a vast number of compounds.
- LO2: Distinguish saturated from unsaturated carbon compounds by identifying the types of C-C bonds present.
- LO3: Classify hydrocarbons into alkanes, alkenes, alkynes, and aromatic (benzene) based on bond types and structure.
- LO4: Describe the typical physical properties of carbon compounds and relate them to intermolecular forces.
- LO5: Summarize the significance of Wöhler's synthesis of urea in the history of organic chemistry.

3. Prerequisite Knowledge

- Understanding of covalent bonding and the concept of valence electrons.
- Knowledge of carbon's atomic number (6) and its electron configuration.
- Familiarity with chemical formulas and bond notation (single, double, triple bonds).

4. Key Terms

Term	Definition
Tetravalency	The ability of carbon to form four covalent bonds because it has four valence electrons.
Catenation	The property of carbon atoms to bond with each other, forming chains, branched structures, or rings.
Saturated hydrocarbon	A hydrocarbon containing only single C-C bonds; also called an alkane.
Unsaturated hydrocarbon	A hydrocarbon that contains one or more double or triple C-C bonds; includes alkenes and alkynes.
Alkane	A saturated hydrocarbon with the general formula C_nH_{2n+2} .
Alkene	An unsaturated hydrocarbon containing at least one C=C double bond; general formula C_nH_{2n} .

Term	Definition
Alkyne	An unsaturated hydrocarbon containing at least one $\text{C}\equiv\text{C}$ triple bond; general formula $\text{C}_n\text{H}_{2n-2}$.
Benzene	An aromatic hydrocarbon with a six-carbon ring and alternating double bonds, formula C_6H_6 .

5. Key Points

- Carbon can form four covalent bonds and link to other carbon atoms, creating many compounds.
- Saturated compounds contain only single C-C bonds, while unsaturated compounds have double or triple bonds.
- Hydrocarbons are grouped as alkanes, alkenes, alkynes, or aromatic (benzene) based on bond type and structure.
- Carbon compounds generally have low melting/boiling points and are poor electrical conductors.
- Wöhler's 1828 preparation of urea disproved the idea that organic compounds require a 'vital force'.

6. Real-World Applications

Classifying Common Formulas

Using the definitions from the textbook, students identify the hydrocarbon class for given molecular formulas.

Think about: Given the formula C_2H_6 , which class of hydrocarbon does it belong to?

Suggested time: 4 minutes

7. Core Notes

Carbon's Tetravalency and Catenation

- Carbon has an atomic number of 6 and four valence electrons.
- Its tetravalent nature lets it form up to four covalent bonds.
- Catenation is the ability of carbon atoms to bond with each other, producing long chains, branched structures, and rings.
- These two features together explain why carbon gives rise to a vast number of compounds.

Saturated vs. Unsaturated Carbon Compounds

- Saturated hydrocarbons (alkanes) have only single bonds between carbon atoms.
- Unsaturated hydrocarbons contain one or more multiple bonds:
 - One or more double bonds → alkenes.
 - One or more triple bonds → alkynes.
- The presence of multiple bonds reduces the number of hydrogen atoms attached to the carbon skeleton.

Classification of Hydrocarbons

- Hydrocarbons contain only carbon and hydrogen.
- Alkanes - saturated hydrocarbons with only single C-C bonds.
- Alkenes - unsaturated hydrocarbons with at least one C=C double bond.
- Alkynes - unsaturated hydrocarbons with at least one C≡C triple bond.
- Benzene (C₆H₆) - an aromatic hydrocarbon with a cyclic structure and alternating double bonds, shown in Figure 4.10.

Physical Properties of Carbon Compounds

- Table 4.1 shows that most carbon compounds melt and boil at lower temperatures than ionic compounds.
- Weak intermolecular forces (e.g., Van der Waals) account for these low temperatures.
- Because they do not form ions, carbon compounds are largely non-conductors of electricity.
- These properties contrast with the strong ionic bonds and high conductivity of typical ionic substances.

Historical Note: Wöhler's Synthesis of Urea

- Early chemists believed organic compounds could only be produced by living organisms.
- Friedrich Wöhler converted ammonium cyanate, an inorganic salt, into urea, an organic compound.
- This experiment showed that organic molecules can be synthesized without any 'vital force', marking the birth of modern organic chemistry.

8. Activities

Concept-Map of Carbon Compounds (12 min)

Students create a concept map linking carbon's tetravalency, catenation, saturated/unsaturated types, hydrocarbon classes, and physical properties.

Mini-Research on Wöhler's Experiment (13 min)

Students read a short summary of Wöhler's synthesis of urea and write a brief paragraph explaining its impact on the view of organic chemistry.

9. Questions

- Q1. What two characteristic features of carbon give rise to the large number of its compounds?
- Q2. Define a saturated hydrocarbon and provide an example.
- Q3. Which type of carbon-carbon bond is present in alkenes but absent in alkanes?
- Q4. Classify the following compounds: C₂H₄ and C₂H₂.
- Q5. Why do most carbon compounds have lower melting and boiling points than ionic compounds?

